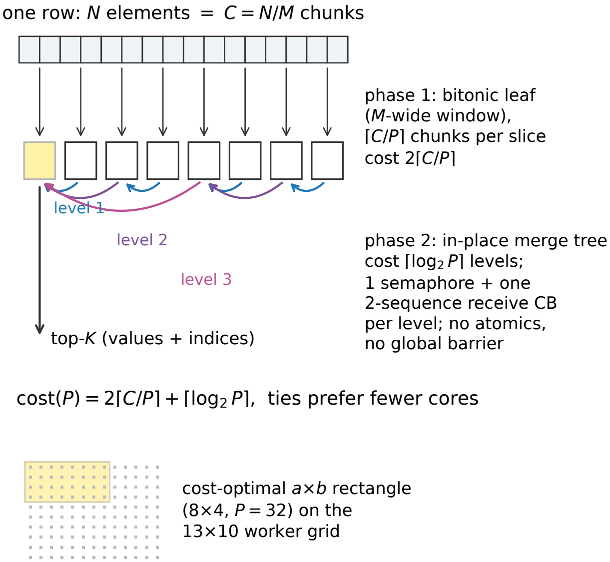
one row: N elements = $C = N/M$ chunks
phase 1: bitonic leaf (M-wide window), $\lceil C/P \rceil$ chunks per slice cost $2\lceil C/P \rceil$
level 1
level 2
level 3
phase 2: in-place merge tree cost $\lceil \log_2 P \rceil$ levels; 1 semaphore + one 2-sequence receive CB per level; no atomics, no global barrier
top-K (values + indices)
$\text{cost}(P) = 2\lceil C/P \rceil + \lceil \log_2 P \rceil$, ties prefer fewer cores
cost-optimal $a \times b$ rectangle (8×4, $P = 32$) on the 13×10 worker grid